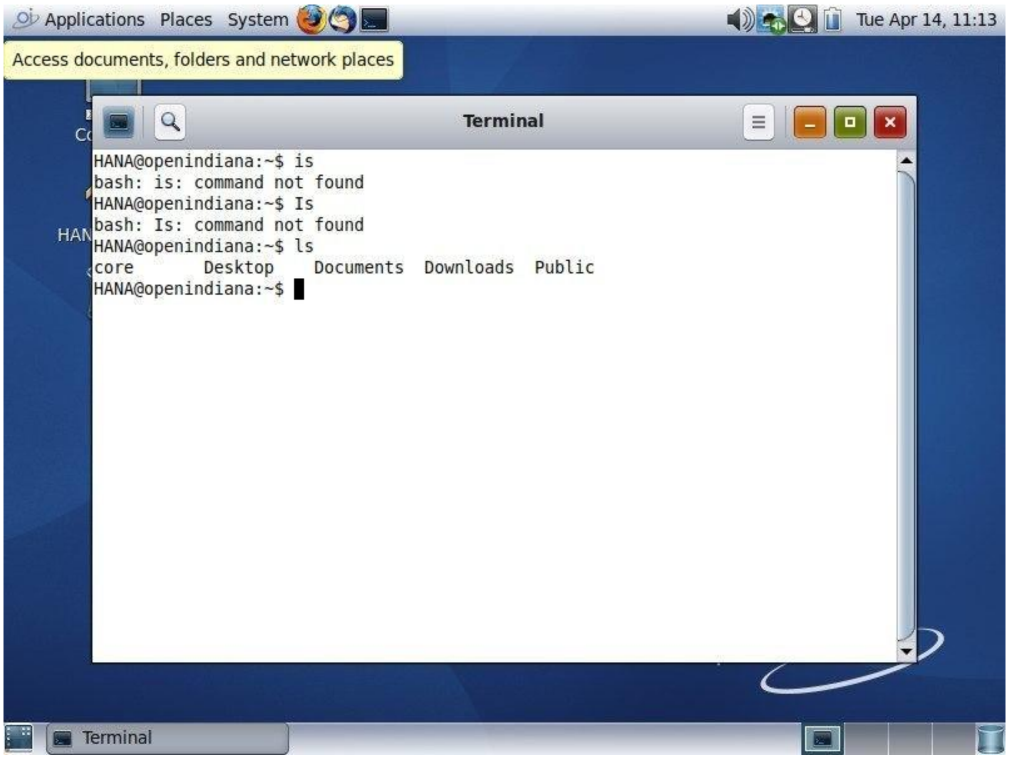
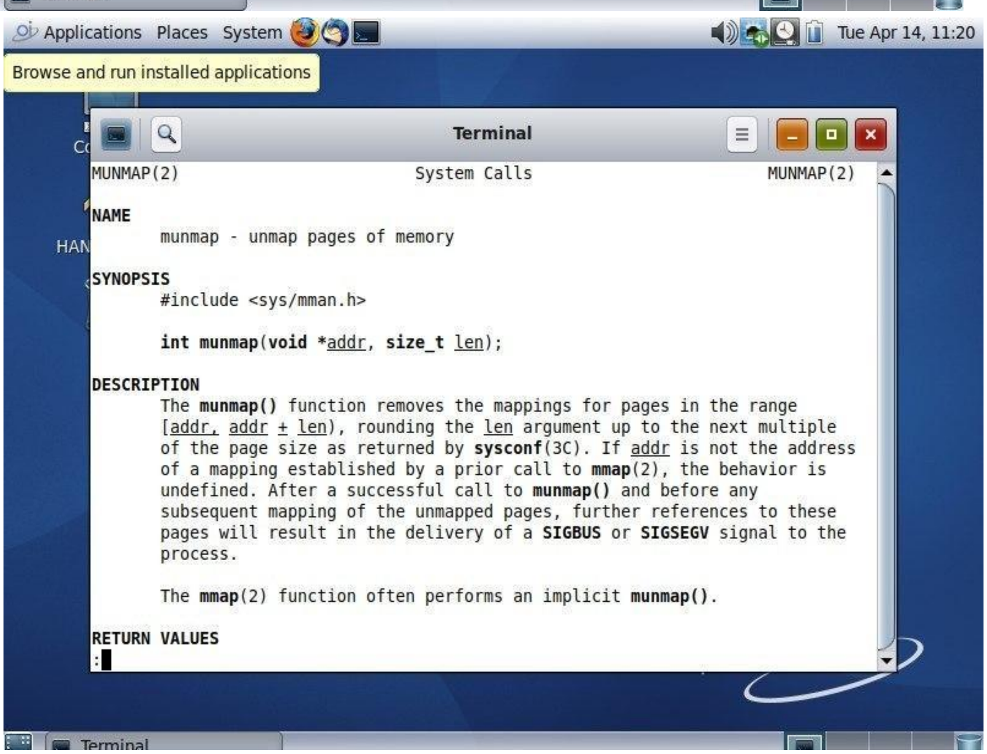
11. System call implmentation

System calls are the interface between user programs and the operating system. In OpenIndiana, system calls allow applications to request services from the kernel, such as file handling, process control, and memory management.

In OpenIndiana, system calls are implemented in a layered way. When a user program wants to perform an operation (for example, reading a file), it does not directly access the hardware. Instead, it sends a request to the operating system through a system call.



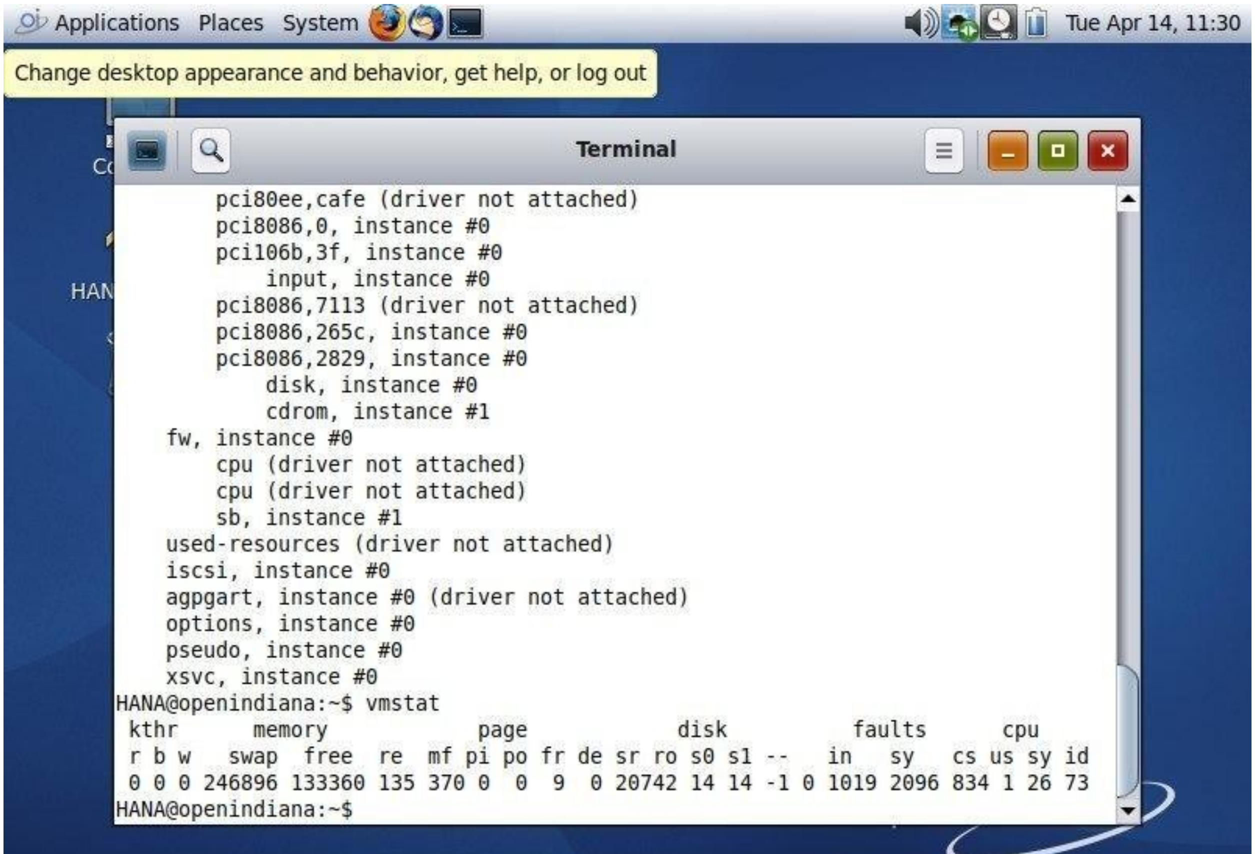


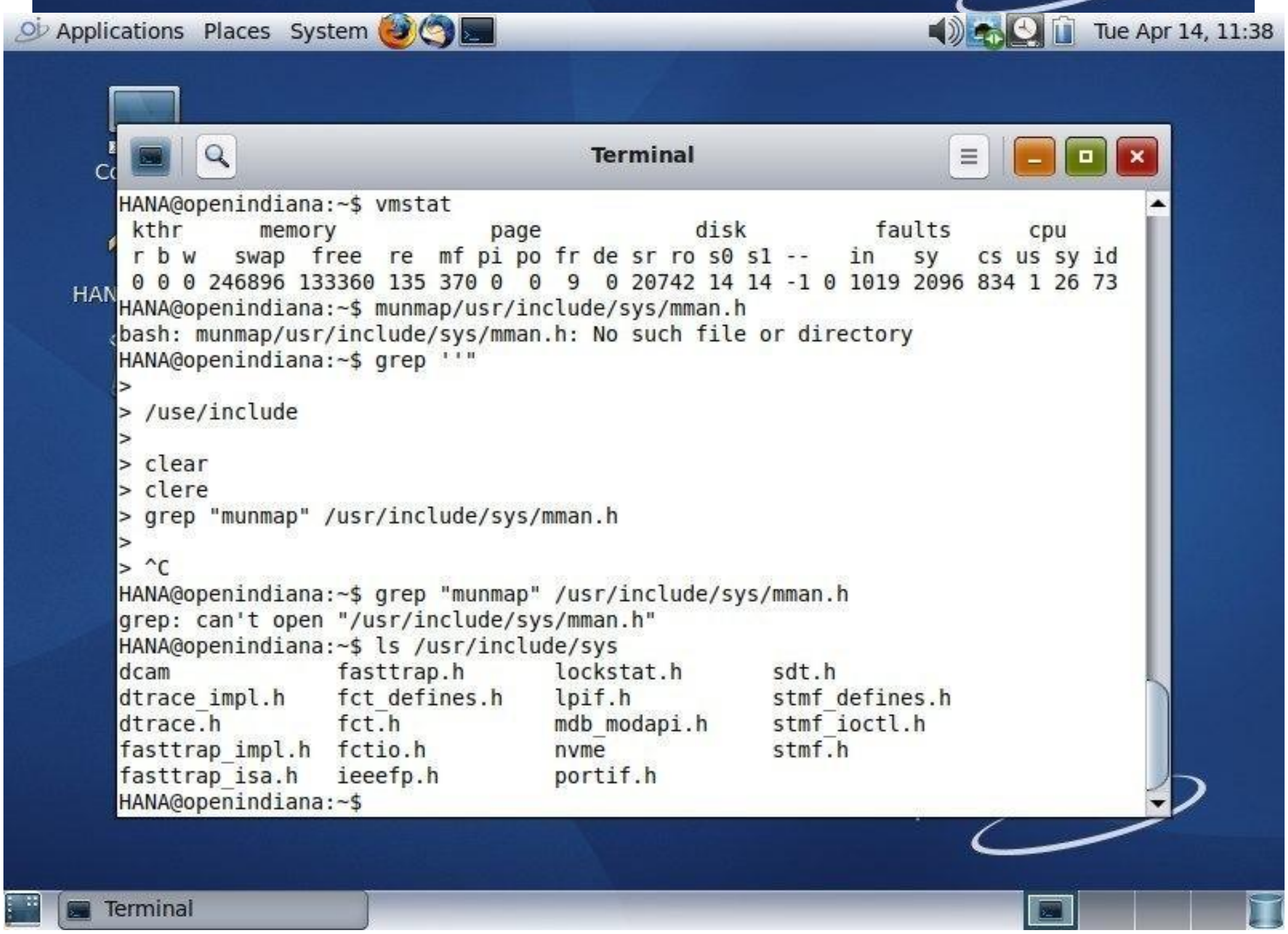
Change desktop appearance and behavior, get help, or log out

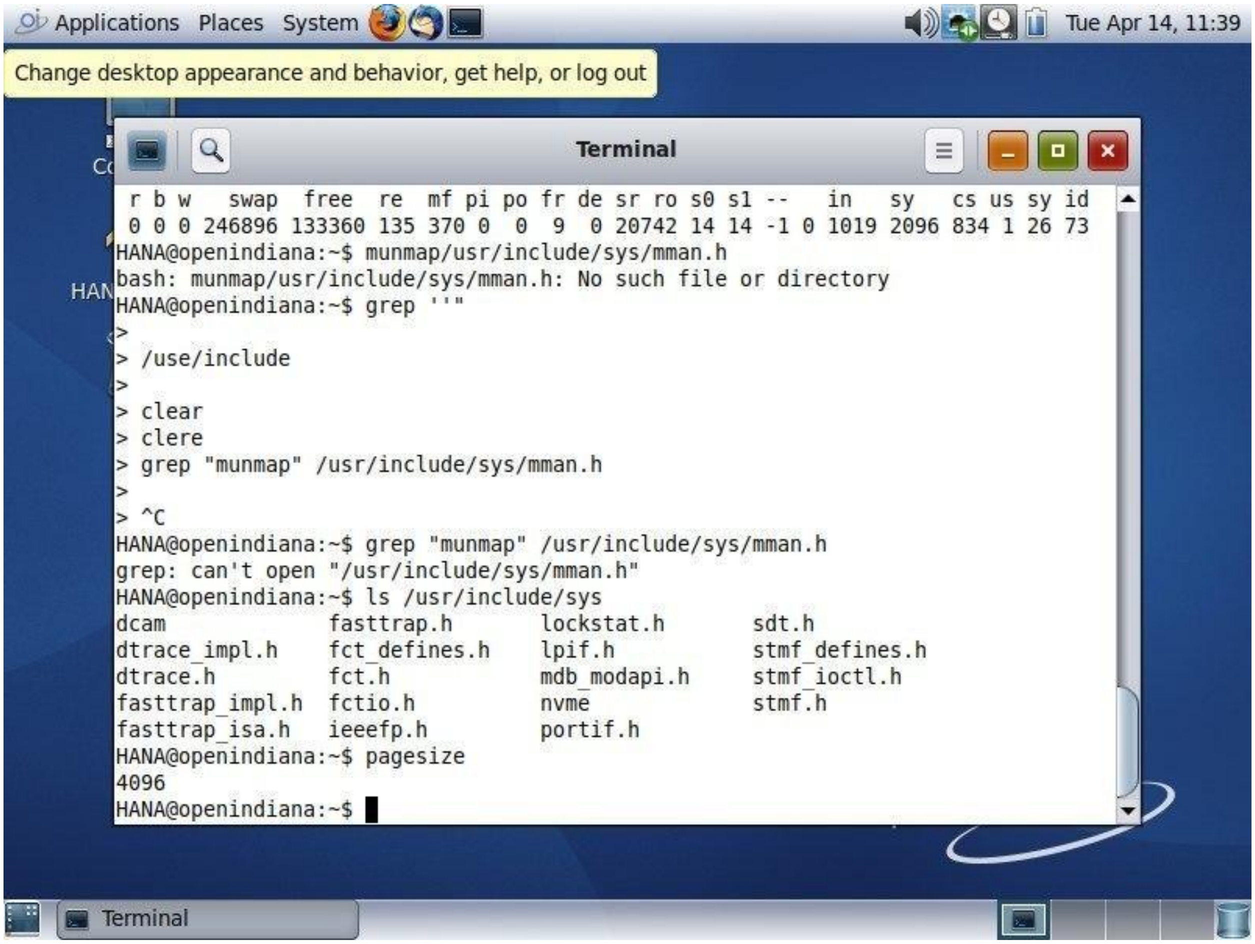
```
            mouse,  1nstance g0
         p1t_beep, Instance â6
      d1spTay, 1nstance #D
      pc18086, be, Instance #B
      pci BDee, cafe (dr1ver not attached)
      pc18686,8, 1nstance GB
      pcilB6b,3f, instance #B
         input, instance #0
      pciB0B6,7l13 (driver not attached)
      pci808b.265c, instance #0
      pciBoB6,2B29, instance #o
         disk, instance #0
         cdrom, 1nstance II
   fu, 1nstance #6
      cpu (d r1ver not attached)
      cpu (d r1ver not attached)
      sb, Ins tance 61
   useJresources (driver not attached
   iscsi, instance #8
   agpgart, instance80 (driver not attached
   options. instance #O
   pseudo, instance 88
   xsvc, instance #0
NANgopenndana:-$
```

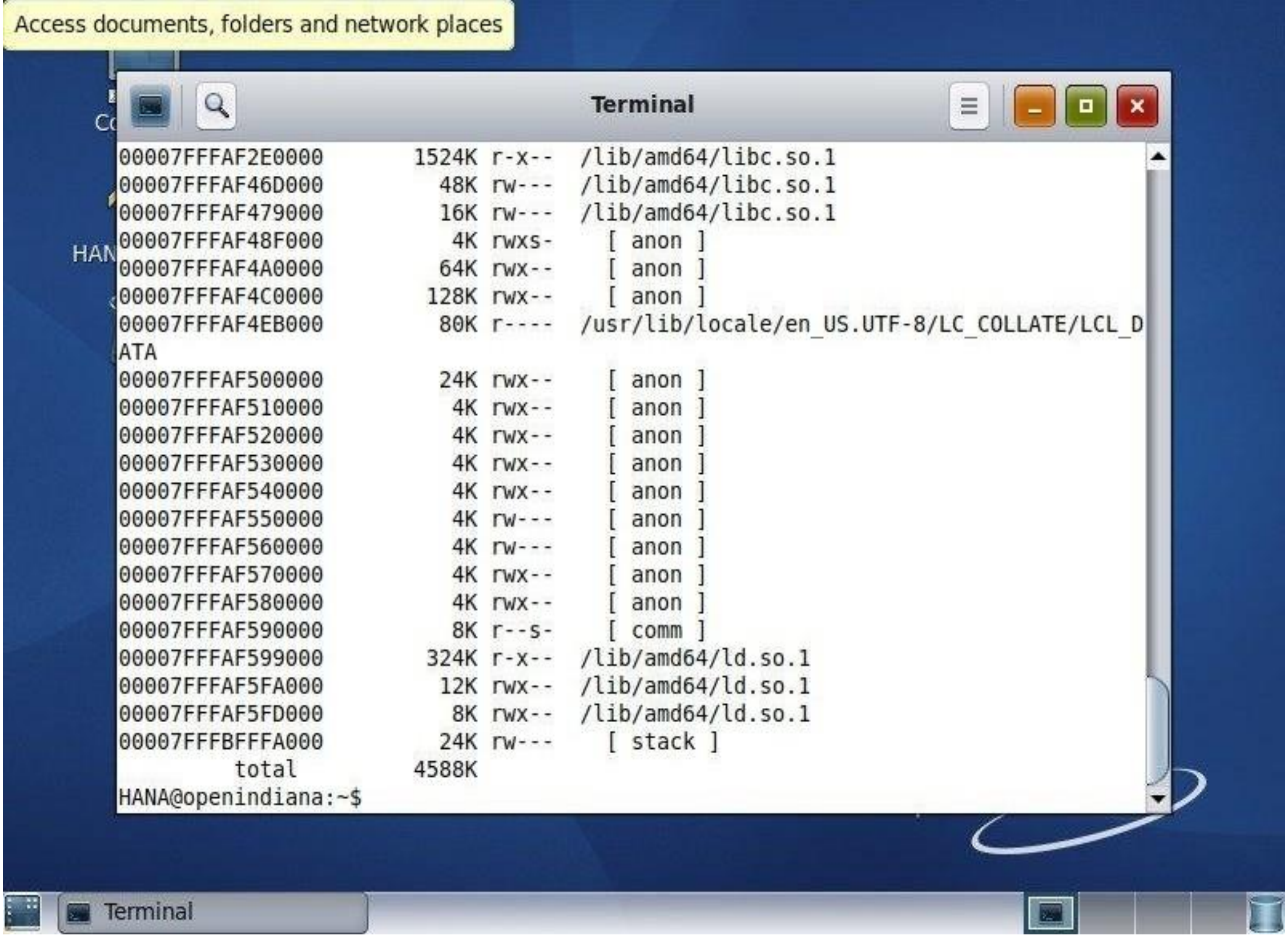
Browse and run installed applications

```
            mouse, 1nstance g0
         p1t beep, Instance #6
      d1sp\ay, 1nstance #o
      pciB0Bb,le, instance #O
      pc18Dee, cafe (dr1ver not attached)
      pciB6B6,8, 1nstance GB
      pet LB6b, 3f, Instance GB
         input , 1nstance gD
      pc18086,7113 (d r1ver not attached)
      pciB0Bb,265c, instance #8
       pc18D86, 2829, Instance #B
          d1sk, Instance #0
         cdrom, 1nstance gI
   fu, Instance #8
      cpu (d r1ver not attached)
      cpu (driver not attached)
      sb, Instance #1
   used-resources (dr1ver not attached)
   1scs1, 1nstance GB
   aqpga rt , 1nstance TO (d r1ver not attached)
   opt1ons , 1nstance g£i
   pseudo, Instance g8
   xsvc , instance #6
HA5Agopen1ndIana:-$ vmstat
```


Applications Places System
Tue Apr 14, 11:30
Change desktop appearance and behavior, get help, or log out
Terminal
pci80ee,cafe (driver not attached)
pci8086,0, instance #0
pci106b,3f, instance #0
input, instance #0
pci8086,7113 (driver not attached)
pci8086,265c, instance #0
pci8086,2829, instance #0
disk, instance #0
cdrom, instance #1
fw, instance #0
cpu (driver not attached)
cpu (driver not attached)
sb, instance #1
used-resources (driver not attached)
iscsi, instance #0
agpgart, instance #0 (driver not attached)
options, instance #0
pseudo, instance #0
xsvc, instance #0
HANA@openindiana:~$ vmstat
kthr memory page disk faults cpu
r b w swap free re mf pi po fr de sr ro s0 s1 -- in sy cs us sy id
0 0 0 246896 133360 135 370 0 0 9 0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$


Applications Places System
Tue Apr 14, 11:38
Terminal
HANA@openindiana:~$ vmstat
kthr memory page disk faults cpu
r b w swap free re mf pi po fr de sr ro s0 s1 -- in sy cs us sy id
0 0 0 246896 133360 135 370 0 0 9 0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$ munmap/usr/include/sys/mman.h
bash: munmap/usr/include/sys/mman.h: No such file or directory
HANA@openindiana:~$ grep '''
>
> /use/include
>
> clear
> clere
> grep "munmap" /usr/include/sys/mman.h
>
> ^C
HANA@openindiana:~$ grep "munmap" /usr/include/sys/mman.h
grep: can't open "/usr/include/sys/mman.h"
HANA@openindiana:~$ ls /usr/include/sys
dcam fasttrap.h lockstat.h sdt.h
dtrace_impl.h fct_defines.h lpif.h stmf_defines.h
dtrace.h fct.h mdb_modapi.h stmf_ioctl.h
fasttrap_impl.h fctio.h nvme stmf.h
fasttrap_isa.h ieeefp.h portif.h
HANA@openindiana:~$
Terminal


Applications Places System
Tue Apr 14, 11:39
Change desktop appearance and behavior, get help, or log out
Terminal
r b w swap free re mf pi po fr de sr ro s0 s1 -- in sy cs us sy id
0 0 0 246896 133360 135 370 0 0 9 0 20742 14 14 -1 0 1019 2096 834 1 26 73
HANA@openindiana:~$ munmap/usr/include/sys/mman.h
bash: munmap/usr/include/sys/mman.h: No such file or directory
HANA@openindiana:~$ grep '''
>
> /use/include
>
> clear
> clere
> grep "munmap" /usr/include/sys/mman.h
>
> ^C
HANA@openindiana:~$ grep "munmap" /usr/include/sys/mman.h
grep: can't open "/usr/include/sys/mman.h"
HANA@openindiana:~$ ls /usr/include/sys
dcam fasttrap.h lockstat.h sdt.h
dtrace_impl.h fct_defines.h lpif.h stmf_defines.h
dtrace.h fct.h mdb_modapi.h stmf_ioctl.h
fasttrap_impl.h fctio.h nvme stmf.h
fasttrap_isa.h ieeefp.h portif.h
HANA@openindiana:~$ pagesize
4096
HANA@openindiana:~$
Terminal



12. Over all personal experiance

This project was about installing and using OpenIndiana in a virtual machine using Oracle VM VirtualBox. It helped to understand how operating systems are installed, configured, and managed in a safe virtual environment.

During the work, I learned basic system setup, virtualization concepts, and how UNIX-based systems operate. I also faced and solved common installation issues, which improved my practical skills.

OpenIndiana uses the ZFS filesystem and follows UNIX standards, making it stable and reliable for learning and system practice.

13. References

- OpenIndiana Official Documentation – https://www.openindiana.org8

- Oracle VM VirtualBox User Guide –
https://www.virtualbox.org/manual8

- UNIX System and POSIX Standards Overview – IEEE Standards Association